

Generation Interconnection System Impact Study Report

For Queue Project Q21-1800

West Huntington 138 kV

December 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1800. The Interconnection Customer proposes a 566.9 MW natural gas combined cycle facility interconnecting at the West Huntington 138 kV substation in the West Virginia area. The request was received on June 27, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 533 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$55,516,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Darrah to Fuel 138 kV L
Contingency: W Huntington to Tecumseh 138 kV
Loading: 316.9% (Rating: 225 MVA)
DFAX: 100.0%
Pre-project loading: 65.0%
2. Facility: W Huntington to Tecumseh 138 kV
Contingency: Fuel to Darrah 138 kV
Loading: 275.0% (Rating: 259 MVA)
DFAX: 100.0%
Pre-project loading: 56.4%
3. Facility: Tecumseh to Krs li Koloa Solar 138 kV
Contingency: Fuel to Darrah 138 kV
Loading: 262.3% (Rating: 267 MVA)
DFAX: 100.0%
Pre-project loading: 50.1%
4. Facility: Darrah to East Huntington 138 kV
Contingency: West Huntington to Tecumseh 138 kV
Loading: 243.8% (Rating: 212 MVA)
DFAX: 100.0%
Pre-project loading: 23.1%
5. Facility: E Huntngton to Ortiz 138 kV
Contingency: West Huntington to Tecumseh 138 kV L
Loading: 230.6% (Rating: 232 MVA)
DFAX: 100.0%
Pre-project loading: 13.5%
6. Facility: Ortiz to W Huntington 138 kV
Contingency: West Huntington to Tecumseh 138 kV
Loading: 223.9% (Rating: 247 MVA)

DFAX: 100.0%

Pre-project loading: 5.5%

7. Facility: Tristate to Krs li Koloa Solar 138 kV L

Contingency: Ceredo Generating Sta to Tristate 138 kV

Loading: 205.5% (Rating: 275 MVA)

DFAX: 55.7%

Pre-project loading: 90.8%

8. Facility: Jefferson Rd to Cedar Cliff 138 kV

Contingency: W Huntington to Tecumseh 138 kV

Loading: 173.4% (Rating: 195 MVA)

DFAX: 40.0%

Pre-project loading: 57.0%

9. Facility: E Proctorville to Jefferson Road 138 kV

Contingency: W Huntington to Tecmseh 138 kV line

Loading: 168.7% (Rating: 191 MVA)

DFAX: 40.0%

Pre-project loading: 49.9%

10. Facility: Zeeland Generating Sta to Cedar Cliff 138 kV

Contingency: West Huntington to Tecumseh 138 kV

Loading: 166.0% (Rating: 209 MVA)

DFAX: 40.0%

Pre-project loading: 57.6%

11. Facility: Zeelnd Generating Station to Mt. View 138 kV L

Contingency: West Huntington to Tecumeh 138 kV

Loading: 160.3% (Rating: 223 MVA)

DFAX: 40.0%

Pre-project loading: 58.6%

12. Facility: Ozark Beach to Mt. View 138 kV

Contingency: W Huntington to Tecumseh 138 kV

Loading: 158.0% (Rating: 234 MVA)

DFAX: 40.0%

Pre-project loading: 61.0%

13. Facility: Ozark Beach to Fuel 138 kV

Contingency: West Huntington to Tecumseh 138 kV

Loading: 157.2% (Rating: 243 MVA)

DFAX: 40.0%

Pre-project loading: 64.0%

14. Facility: Crystal to Eastproctorville 138 kV

Contingency: W Huntington to Tecumseh 138 kV

Loading: 142.5% (Rating: 215 MVA)

DFAX: 40.0%

Pre-project loading: 37.0%

15. Facility: Ceredo Generating Sta to Tristate 138 kV L

Contingency: Krs li Koloa Solar Substation to Tristate 138 kV Line

Loading: 140.5% (Rating: 403 MVA)

DFAX: 45.2%

Pre-project loading: 77.0%

16. Facility: Fuel to Moses Lake 138 kV

Contingency: W Huntington to Tecumseh 138 kV

Loading: 130.7% (Rating: 248 MVA)

DFAX: 60.0%

Pre-project loading: 6.6%

17. Facility: 7Th Avenue to Crystal 138 kV line

Contingency: Tecumseh to West Untington 138 kV

Loading: 125.1% (Rating: 233 MVA)

DFAX: 40.0%

Pre-project loading: 27.5%

18. Facility: Moses Lake to Crystal Creamery 138 kV Line

Contingency: Tecumseh to W Huntington 138 kV line

Loading: 120.1% (Rating: 259 MVA)

DFAX: 60.0%

Pre-project loading: 11.2%

19. Facility: Barick to Crystal Creamery 138 kV

Contingency: W Huntington to Tecumseh 138 kV

Loading: 118.6% (Rating: 260 MVA)

DFAX: 60.0%

Pre-project loading: 12.3%

20. Facility: Sneath Lane to Barrick Substation 138 kV

Contingency: Tecumseh to W Huntington 138 kV

Loading: 116.5% (Rating: 262 MVA)

DFAX: 60.0%

Pre-project loading: 13.3%

21. Facility: Sneath Lane to Queen Avenue 138 kV

Contingency: Tecumseh Substation to West Huntington 138 kV

Loading: 108.9% (Rating: 269 MVA)

DFAX: 60.0%

Pre-project loading: 17.6%

22. Facility: Cannon Beach to Queen Avenue 138 kV L

Contingency: W Huntington to Tecumseh 138 kV

Loading: 107.1% (Rating: 271 MVA)

DFAX: 60.0%

Pre-project loading: 18.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Darrah to Fuel 138 kV L	Rebuild line with higher-capacity conductor	\$4,605,000	73% (\$3,355,000)
W Huntington to Tecumseh 138 kV	Rebuild line with higher-capacity conductor	\$2,367,000	73% (\$1,725,000)
Tecumseh to Krs li Koloa Solar 138 kV	Rebuild line with higher-capacity conductor	\$1,999,000	73% (\$1,456,000)
Darrah to E Huntington 138 kV	Rebuild line with higher-capacity conductor	\$2,817,000	100% (\$2,817,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Huntington to Ortiz 138 kV	Rebuild line with higher-capacity conductor	\$3,046,000	100% (\$3,046,000)
Ortiz to W Huntington 138 kV	Rebuild line with higher-capacity conductor	\$1,592,000	100% (\$1,592,000)
Tristate to Krs li Koloa Solar 138 kV L	Rebuild line with higher-capacity conductor	\$3,952,000	47% (\$1,869,000)
Jefferson Road to Cedar Cliff 138 kV L	Rebuild line with higher-capacity conductor	\$12,396,000	73% (\$9,033,000)
E Proctrville Substation to Jeferson Rd 138 kV Line	Rebuild line with higher-capacity conductor	\$1,992,000	73% (\$1,451,000)
Zeeland Generating Sta to Cedar Cliff 138 kV	Rebuild line with higher-capacity conductor	\$3,612,000	73% (\$2,632,000)
Zeelnd Generating Station to Mt. View 138 kV L	Rebuild line with higher-capacity conductor	\$4,225,000	73% (\$3,078,000)
Ozark Beach to Mt. View 138 kV	Rebuild line with higher-capacity conductor	\$6,090,000	73% (\$4,438,000)
Ozark Beach to Fuel 138 kV	Rebuild line with higher-capacity conductor	\$5,371,000	73% (\$3,913,000)
E Proctorville to Crystal 138 kV	Rebuild line with higher-capacity conductor	\$1,470,000	73% (\$1,071,000)
Trisate to Ceredo Generating Sta 138 kV Line	Rebuild line with higher-capacity conductor	\$1,857,000	39% (\$732,000)
Moses Lake Substation to Fuel 138 kV L	Rebuild line with higher-capacity conductor	\$3,060,000	73% (\$2,230,000)
Crystal to 7Th Avenue 138 kV	Rebuild line with higher-capacity conductor	\$4,717,000	73% (\$3,437,000)
Moses Lake to Crystal Creamery 138 kV Line	Rebuild line with higher-capacity conductor	\$2,965,000	73% (\$2,161,000)
Barick to Crystal Creamery 138 kV	Rebuild line with higher-capacity conductor	\$2,033,000	73% (\$1,482,000)
Barrick to Sneath Lane 138 kV line	Rebuild line with higher-capacity conductor	\$2,169,000	73% (\$1,580,000)
Sneath Lane to Queen Avenue 138 kV	Reconductor line	\$2,315,000	73% (\$1,687,000)
Cannon Beach to Queen Avenue 138 kV L	Reconductor line	\$1,003,000	73% (\$731,000)
	Total allocated to project		\$55,516,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.